

Operating Systems Comprehensive Study Guide

MEMORY MANAGEMENT, DISK SCHEDULING, PROCESS MANAGEMENT & SYNCHRONIZATION

Part 1: Memory Management & Disk Scheduling

1. Define Logical Address and Physical Address.

Logical Address: The address generated by the CPU during program execution. It is also called a virtual address.

Physical Address: The actual address in the main memory (RAM) where data is stored.

2. What is a Relocation Register? Why is it used?

A Relocation Register contains the starting physical address of a process in memory.

Use: It helps convert logical addresses into physical addresses, allowing programs to be loaded anywhere in memory without modifying their code.

3. What is a Memory Management Unit (MMU)?

The Memory Management Unit (MMU) is a hardware component that translates logical addresses generated by the CPU into physical addresses in memory and provides memory protection.

4. What is Virtual Memory?

Virtual Memory is a memory management technique that allows a process to execute even if it is not completely loaded into main memory. It uses secondary storage as an extension of RAM.

5. Define Swapping.

Swapping is a memory management technique in which a process is temporarily moved from main memory to secondary storage and later brought back into memory for execution.

6. What is Memory Protection?

Memory Protection is a mechanism that prevents one process from accessing or modifying the memory allocated to another process, ensuring system security and stability.

7. What are First-Fit, Best-Fit, and Worst-Fit Memory Allocation?

First-Fit: Allocates the first free memory block that is large enough for the process.

Best-Fit: Allocates the smallest free block that can accommodate the process.

Worst-Fit: Allocates the largest available free memory block to the process.

8. What is Fragmentation? List its Types.

Fragmentation is the wastage of memory caused by inefficient memory allocation.

Types:

1. Internal Fragmentation – Unused space inside an allocated memory block.
2. External Fragmentation – Free memory exists but is scattered into small non-contiguous blocks.

9. What is Paging and Page Fault?

Paging: A memory management technique that divides logical memory into fixed-size pages and physical memory into frames.

Page Fault: An interrupt that occurs when a required page is not present in main memory and must be loaded from secondary storage.

10. What is Segmentation?

Segmentation is a memory management technique in which a program is divided into logical units called segments (such as code, data, and stack). Each segment has its own size and address.

11. What is Segmentation Hardware?

Segmentation Hardware is the hardware support used to implement segmentation. It consists of a segment table, segment-table base register (STBR), and segment-table length register (STLR) to translate logical addresses into physical addresses.

12. What is Demand Paging?

Demand Paging is a virtual memory technique in which pages are loaded into main memory only when they are needed for execution.

13. What is Pure Demand Paging?

Pure Demand Paging is a paging scheme where no page is loaded into memory initially. A page is brought into memory only when a page fault occurs.

14. List Two Hardware Supports for Demand Paging.

15. Page Table

16. Translation Lookaside Buffer (TLB)

(Some texts also mention MMU and valid/invalid bit support.)

15. What are the Major Problems with Implementing Demand Paging?

16. Page Fault Overhead – Accessing a missing page causes delays.

17. Thrashing – Excessive page faults reduce system performance.

16. Define Thrashing and Virtual Memory.

Thrashing: A condition in which the system spends more time swapping pages between memory and disk than executing processes.

Virtual Memory: A memory management technique that uses secondary storage to extend the available main memory.

17. Differentiate Between Logical and Physical Address Space.

Logical Address Space	Physical Address Space
Generated by CPU	Actual address in RAM
Also called virtual address	Also called real address
Seen by the user/program	Seen by memory hardware

18. Differentiate Between Paging and Segmentation.

Paging	Segmentation
Divides memory into fixed-size pages	Divides memory into variable-size segments
Eliminates external fragmentation	May suffer from external fragmentation
Programmer is unaware of pages	Programmer is aware of segments

19. What are the Limitations of Swapping?

- 20. Swapping is time-consuming due to disk I/O operations.
- 21. It reduces system performance when large processes are swapped frequently.
- 22. Requires sufficient backing store (secondary storage).

20. Differentiate Between Swapping and Demand Paging.

Swapping	Demand Paging
Entire process is moved between memory and disk	Only required pages are loaded into memory
Higher memory transfer overhead	Lower memory transfer overhead
Less efficient	More efficient and widely used

21. What is the Basic Approach of Page Replacement?

Page Replacement is the process of selecting a page in memory to remove when a page fault occurs and no free frame is available, so that the required page can be loaded.

22. What is FIFO Page Replacement Algorithm?

FIFO (First-In First-Out) replaces the page that has been in memory for the longest time. The oldest page is removed first.

23. What is Optimal Page Replacement Algorithm?

Optimal Page Replacement replaces the page that will not be used for the longest period in the future. It produces the minimum number of page faults.

24. What is LRU Page Replacement Algorithm?

LRU (Least Recently Used) replaces the page that has not been used for the longest period of time in the past.

25. What Do You Mean by Belady's Anomaly?

Belady's Anomaly is a situation in which increasing the number of page frames results in an increase in the number of page faults, usually in the FIFO page replacement algorithm.

26. Define Working-Set Model.

The Working-Set Model is a memory management technique that keeps track of the set of pages currently being used by a process to prevent thrashing.

27. What is Page Fault Frequency (PFF)?

Page Fault Frequency (PFF) is a method that controls the page fault rate of a process by adjusting the number of frames allocated to it.

28. What is Disk Scheduling?

Disk Scheduling is the process of deciding the order in which disk I/O requests are serviced to improve disk access efficiency and reduce seek time.

29. What is FCFS Scheduling?

FCFS (First-Come, First-Served) is a disk scheduling algorithm that serves disk requests in the order they arrive.

30. What is SSTF Scheduling?

SSTF (Shortest Seek Time First) selects the disk request that is closest to the current head position, thereby reducing seek time.

31. What is SCAN Scheduling?

SCAN Scheduling moves the disk head in one direction, servicing all requests along the way, and then reverses direction. It is also called the Elevator Algorithm.

32. What is C-SCAN Scheduling?

C-SCAN (Circular SCAN) moves the disk head in one direction only. After reaching the end, it returns to the beginning without servicing requests on the return trip.

33. What is LOOK Scheduling?

LOOK Scheduling is similar to SCAN, but the disk head moves only as far as the last request in each direction before reversing, reducing unnecessary movement.

Part 2: Process Management & Scheduling

1. What is a Process? List the Four Components of a Process.

A process is a program that is currently being executed.

Four components of a process:

1. Program Code (Text Section)
2. Data Section
3. Stack
4. Heap

2. Differentiate Between Process and Program.

Program	Process
A passive set of instructions stored on disk	A program in execution
Static in nature	Dynamic in nature
Does not require resources while stored	Requires CPU, memory, and I/O resources
Not an active entity	Active entity

3. List Out Any Four Process States.

1. New
2. Ready
3. Running
4. Waiting (Blocked)

4. What is PCB? List Its Components.

PCB (Process Control Block) is a data structure used by the operating system to store information about a process.

Components:

1. Process ID (PID)
2. Process State
3. Program Counter
4. CPU Registers
5. Memory Management Information
6. I/O Status Information

5. What is Context Switch?

A context switch is the process of saving the state of a currently running process and loading the state of another process so that CPU execution can be transferred.

6. Define Thread.

A thread is the smallest unit of CPU execution within a process. Multiple threads can exist within a single process and share its resources.

7. Define Dispatcher.

A dispatcher is the operating system component that transfers CPU control from one process to another selected by the scheduler.

8. Define Independent and Cooperating Processes.

Independent Process: A process that cannot affect or be affected by other processes.

Cooperating Process: A process that can affect or be affected by other processes and may share data with them.

9. What Do You Mean by Inter-Process Communication (IPC)?

Inter-Process Communication (IPC) is a mechanism that allows processes to exchange data and synchronize their activities.

10. What is a Scheduler? Mention Its Types.

A scheduler is an operating system component that selects processes for execution.

Types of Schedulers:

1. Long-Term Scheduler
2. Short-Term Scheduler
3. Medium-Term Scheduler

11. What is Ready Queue and Device Queue?

Ready Queue: A queue that contains processes waiting for CPU allocation.

Device Queue: A queue that contains processes waiting for access to a specific I/O device.

12. Define CPU Burst and I/O Burst.

CPU Burst: The period during which a process is executing instructions on the CPU.

I/O Burst: The period during which a process performs input/output operations and waits for I/O completion.

13. Define Throughput and Turnaround Time.

Throughput: The number of processes completed by the system per unit time.

Turnaround Time: The total time taken from process submission to process completion.

Turnaround Time = Completion Time – Arrival Time

14. Define Response Time and Arrival Time.

Response Time: The time between the submission of a process and the first response produced by the system.

Arrival Time: The time at which a process enters the ready queue and becomes available for execution.

15. What is Waiting Time?

Waiting Time is the total time a process spends waiting in the ready queue before getting CPU time.

Waiting Time = Turnaround Time – CPU Burst Time

16. Define Preemptive Scheduling and Non-Preemptive Scheduling.

Preemptive Scheduling: The CPU can be taken away from a running process and assigned to another process.

Non-Preemptive Scheduling: Once a process gets the CPU, it continues execution until it completes or enters the waiting state.

17. Difference Between Job and Process.

Job	Process
A program waiting for execution	A program currently in execution
Stored on secondary storage	Loaded into main memory
Passive entity	Active entity
Does not use CPU	Uses CPU and other resources

18. What is Aging?

Aging is a technique used to prevent starvation by gradually increasing the priority of processes that wait in the ready queue for a long time.

19. List the Disadvantages of SJF.

1. Difficult to predict the length of the next CPU burst.
2. Long processes may suffer from starvation.
3. Not suitable for interactive systems.
4. Implementation is complex due to burst-time estimation.

20. Differentiate Long-Term Scheduler and Short-Term Scheduler.

Long-Term Scheduler	Short-Term Scheduler
Selects jobs from the job pool	Selects processes from the ready queue
Controls the degree of multiprogramming	Allocates CPU to processes
Invoked less frequently	Invoked very frequently
Slower in operation	Faster in operation

Part 3: Process Synchronization & Deadlocks

1. Define Critical Section Problem.

The Critical Section Problem is the problem of designing a protocol that ensures that when one process is executing in its critical section, no other process is allowed to execute in its critical section at the same time.

2. What are the Requirements of the Critical Section Problem?

A solution to the critical section problem must satisfy the following requirements:

1. Mutual Exclusion – Only one process can execute in its critical section at a time.
2. Progress – If no process is in the critical section, the selection of the next process cannot be postponed indefinitely.
3. Bounded Waiting – A process should not wait indefinitely to enter its critical section.

3. Write the General Structure of a Typical Critical Section Problem.

```
do {  
    Entry Section  
  
    Critical Section  
  
    Exit Section  
  
    Remainder Section  
  
} while (true);
```

Entry Section: Requests permission to enter the critical section.

Critical Section: Accesses shared resources.

Exit Section: Releases the resource.

Remainder Section: Executes the remaining code.

4. What is a Semaphore?

A Semaphore is a synchronization tool used to control access to shared resources by multiple processes. It is an integer variable that can be modified only through the wait() and signal() operations.

5. Write the General Structure of wait() and signal().

```
wait(S)
{
    while (S <= 0);
    S--;
}

signal(S)
{
    S++;
}
```

Where S is a semaphore variable.

6. What is Deadlock? Give a Real-Life Example.

A Deadlock is a situation in which two or more processes are waiting indefinitely for resources held by each other, and none of them can proceed.

Real-Life Example: Two vehicles meet on a narrow bridge from opposite directions. Neither vehicle can move forward unless the other moves back, causing a deadlock.

7. Which are the Necessary Conditions for Deadlock?

The four necessary conditions for deadlock are:

1. Mutual Exclusion – At least one resource must be non-shareable.
2. Hold and Wait – A process holds one resource while waiting for another.
3. No Preemption – Resources cannot be forcibly taken away.
4. Circular Wait – A circular chain of processes exists where each process waits for a resource held by the next process.

8. What is Resource-Allocation Graph?

A Resource-Allocation Graph (RAG) is a directed graph used to represent the allocation of resources to processes and the requests made by processes for resources. It helps in analyzing and detecting deadlocks.

9. Mention the Methods for Handling Deadlocks.

1. Deadlock Prevention
2. Deadlock Avoidance
3. Deadlock Detection and Recovery
4. Ignore the Deadlock Problem (Ostrich Approach)

10. Explain the Term Safe State.

A Safe State is a state in which the system can allocate resources to all processes in some order and avoid deadlock.

11. What is Race Condition?

A Race Condition occurs when two or more processes access and modify shared data simultaneously, and the final result depends on the order of execution.

12. What is Deadlock Prevention?

Deadlock Prevention is a method of ensuring that at least one of the necessary conditions for deadlock never occurs, thereby preventing deadlocks from happening.

13. Mention Two Ways of Recovery from Deadlock.

1. Process Termination
2. Resource Preemption

14. Mention Different Ways to Eliminate Resource Preemption.

1. If a process requests a resource that cannot be allocated immediately, all resources currently held by the process are released.
2. Resources are preempted from a process and allocated to another process until the required resources become available.

15. Mention Different Ways to Eliminate Deadlock by Process Termination.

1. Abort all deadlocked processes.
2. Abort one process at a time until the deadlock cycle is eliminated.